Does Tokenization Fragmentation Predict Miscalibration in Bangla LLMs? A Cross-Lingual Study

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Abstract.

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1 Introduction

1.1 Background and Motivation

1.2 Problem Statement

1.3 Research Gap

1.4 Research Questions and Contributions

- **RQ1:** How does tokenization fragmentation differ between Bangla and English under the selected model tokenizers?
- **RQ2:** Is item-level tokenization fragmentation associated with confidence and calibration behaviour on Bangla evaluation items?
- **RQ3:** Does this association vary across models, subjects, or language conditions?

1.5 Paper Organization

2 Related Work

2.1 Subword Tokenization and Multilingual Tokenizer Behaviour

2.2 Measuring Tokenization Fragmentation

2.3 Bangla LLMs and Bangla Benchmarking

2.4 Confidence Estimation and Calibration in LLMs

2.5 Summary of the Gap

3 Study Design and Research Questions

3.1 Study Overview

3.2 Hypotheses

4 Models and Datasets

4.1 Dataset Sources and Versions

4.2 Bangla and English Evaluation Conditions

4.3 Question Pairing and Translation Quality

4.4 Model and Tokenizer Selection

4.5 Data Cleaning and Preprocessing

4.6 Ethics and Data Handling

5 Tokenization Fragmentation Measurement

5.1 Text Unit Being Tokenized

5.2 Tokenizer Execution and Reproducibility

5.3 Word Counting and Normalization

5.4 Tokens per Word (TPW)

$$\mathrm{TPW}(x) = \frac{N_{\mathrm{tokens}}(x)}{N_{\mathrm{words}}(x)}. \tag{1}$$

5.5 Subword Token Retention Ratio (STRR)

5.6 Worked Tokenization Examples

6 Confidence and Calibration Methodology

6.1 Prompt and Multiple-Choice Scoring Protocol

6.2 Confidence Definition

6.3 Accuracy

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(\hat{y}_i = y_i). \tag{2}$$

6.4 Brier Score

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}(p_{ik} - y_{ik})^2. \tag{3}$$

6.5 Expected Calibration Error (ECE)

$$\text{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{N} \left|\text{acc}(I_b) - \text{conf}(I_b)\right|. \tag{4}$$

6.6 Reliability Diagrams

6.7 Calibration Variants and Data Leakage Prevention

7 Experimental Setup

7.1 Evaluation Pipeline

7.2 Primary and Secondary Outcomes

7.3 Statistical Analysis Plan

7.4 Robustness and Sensitivity Checks

8 Results

8.1 Dataset and Model Coverage

8.2 Tokenization Fragmentation by Language and Model

8.3 Accuracy and Calibration Performance

8.4 Association Between Fragmentation and Calibration

8.5 Robustness Checks

9 Discussion

9.1 Summary of Main Findings

9.2 Interpretation and Comparison with Prior Work

9.3 Implications for Bangla LLM Evaluation

10 Limitations and Threats to Validity

10.1 Construct Validity

10.2 Internal Validity

10.3 External Validity

10.4 Statistical Conclusion Validity

11 Conclusion and Future Work

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References